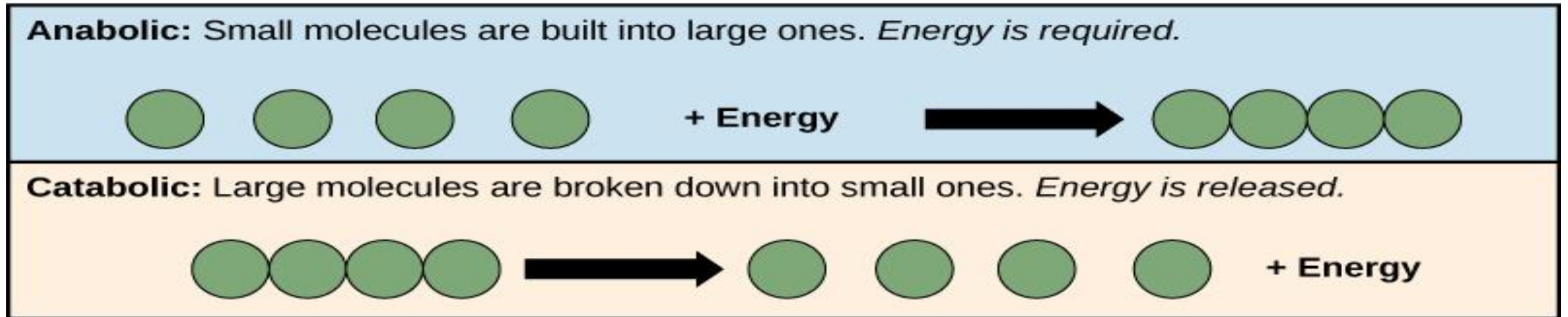


Metabolism

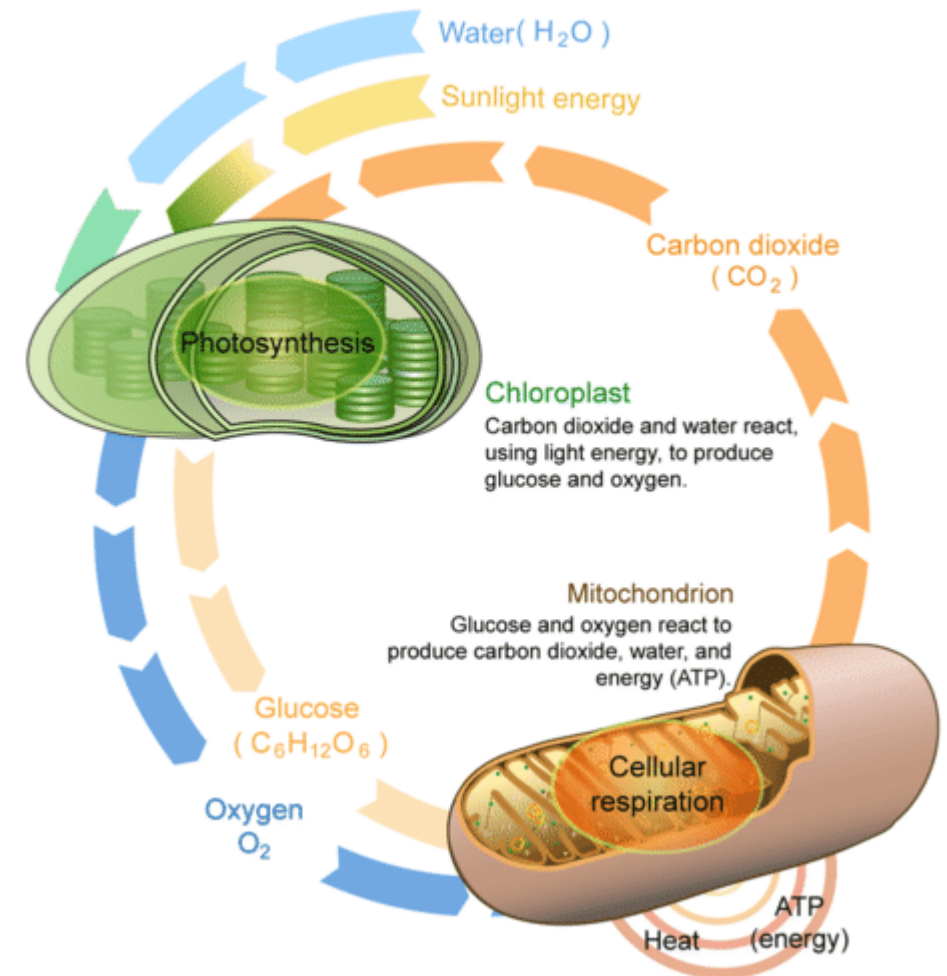
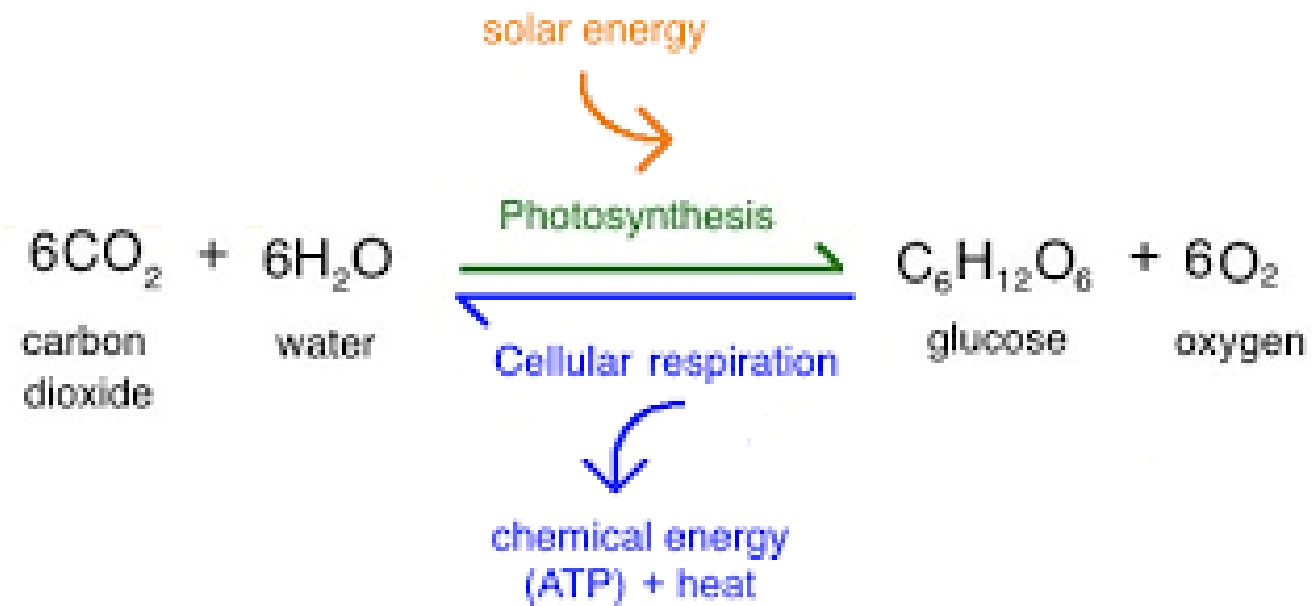
- Metabolism = represents all the chemical reactions and processes taking place inside your cells resulting in LIFE.
- Catabolism = chemical reactions resulting in breaking down big molecules
 - releases energy
- Anabolism = chemical reactions resulting in building up molecules
 - requires energy
- Metabolic pathway = all the consecutive reactions resulting in a specific product
 - Examples:.....

Anabolism vs. Catabolism

Metabolic pathways

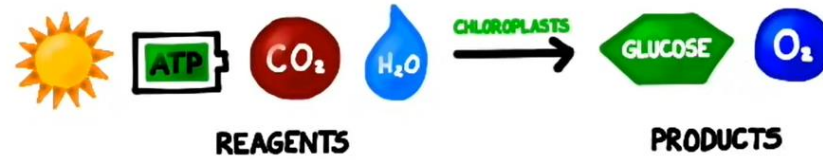


- Catabolic pathways are those that generate energy by breaking down larger molecules. Anabolic pathways are those that require energy to synthesize larger molecules. Both types of pathways are required for maintaining the cell's energy balance.

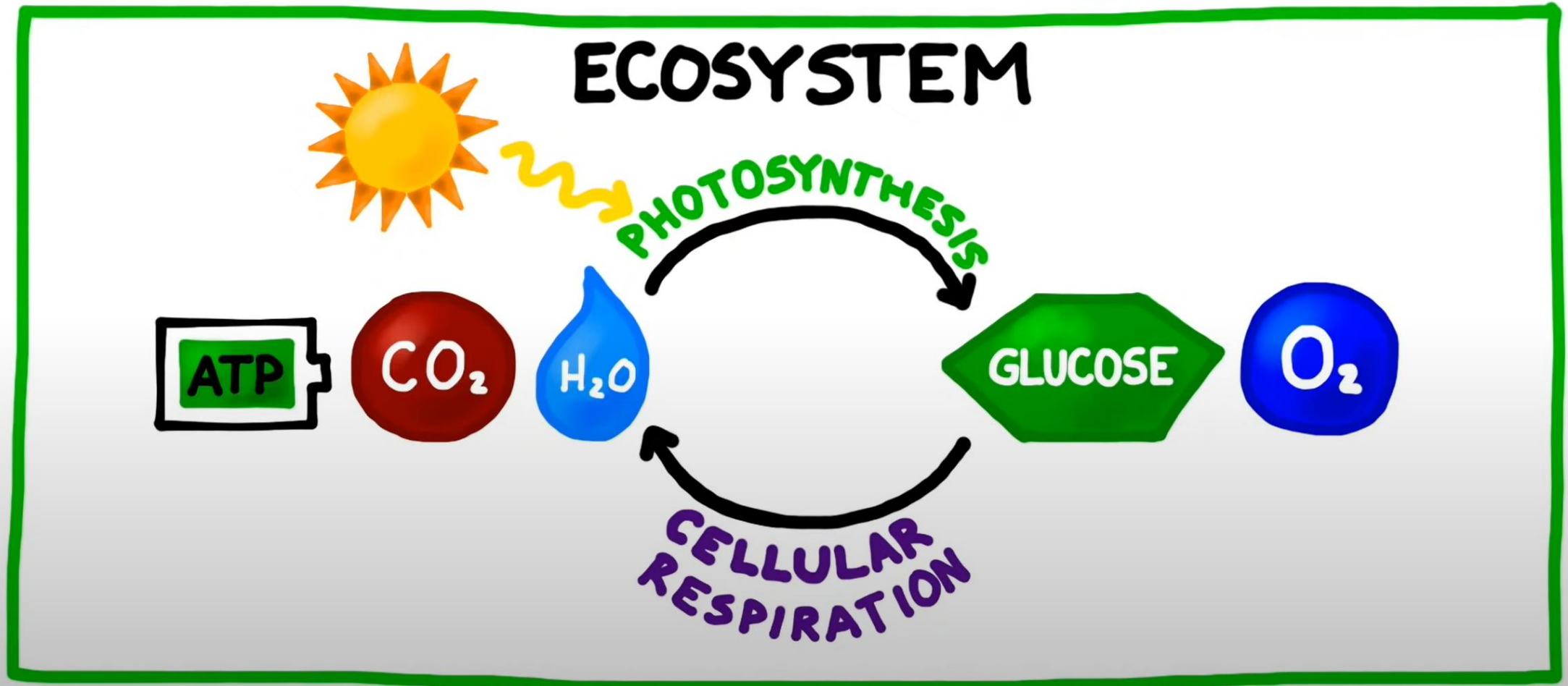


PHOTOSYNTHESIS

REAGENTS +
PRODUCTS:



CELLULAR RESPIRATION



PHOTOSYNTHESIS

LOCATION:

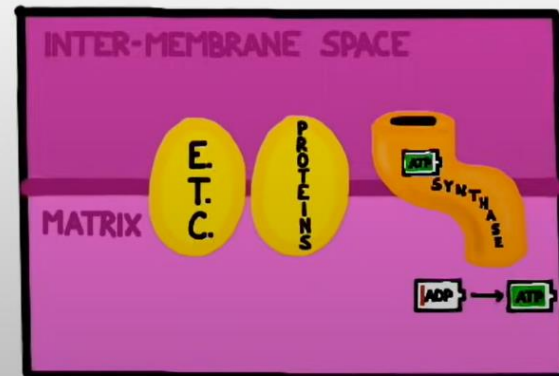
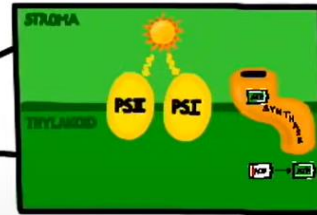
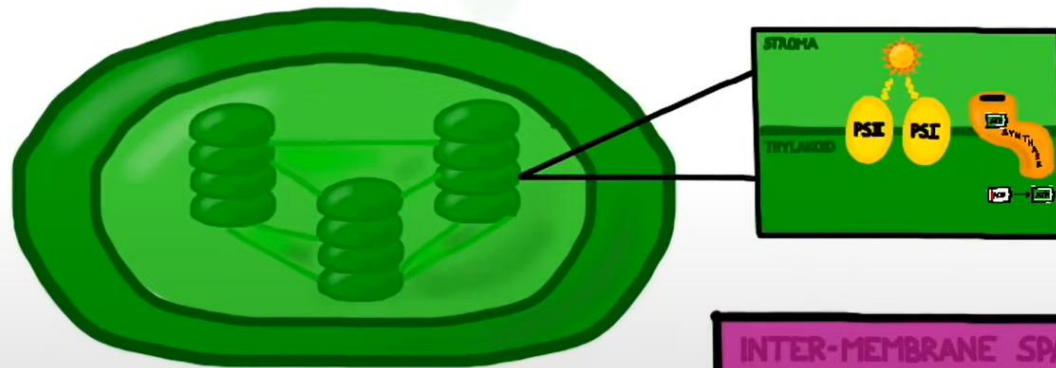
CHLOROPLASTS

INNER
COMPARTMENT:

THYLAKOID

OUTER
COMPARTMENT:

STROMA



[Bing Videos:](#)
Photosynthesis

CELLULAR RESPIRATION

CYTOPLASM + MITOCHONDRIA

MATRIX

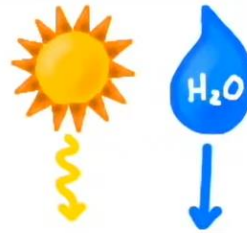
INTER-MEMBRANE SPACE



MAJOR
STEPS:

PHOTOSYNTHESIS

PHOTO:

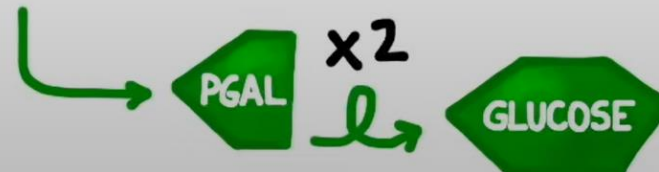


LIGHT DEPENDENT
REACTIONS

SYNTHESIS:



LIGHT
INDEPENDENT
REACTIONS



ATMOSPHERE



CELLULAR RESPIRATION



GLYCOLYSIS



KREB'S
CYCLE
 $\times 2$



ELECTRON
TRANSPORT
CHAIN

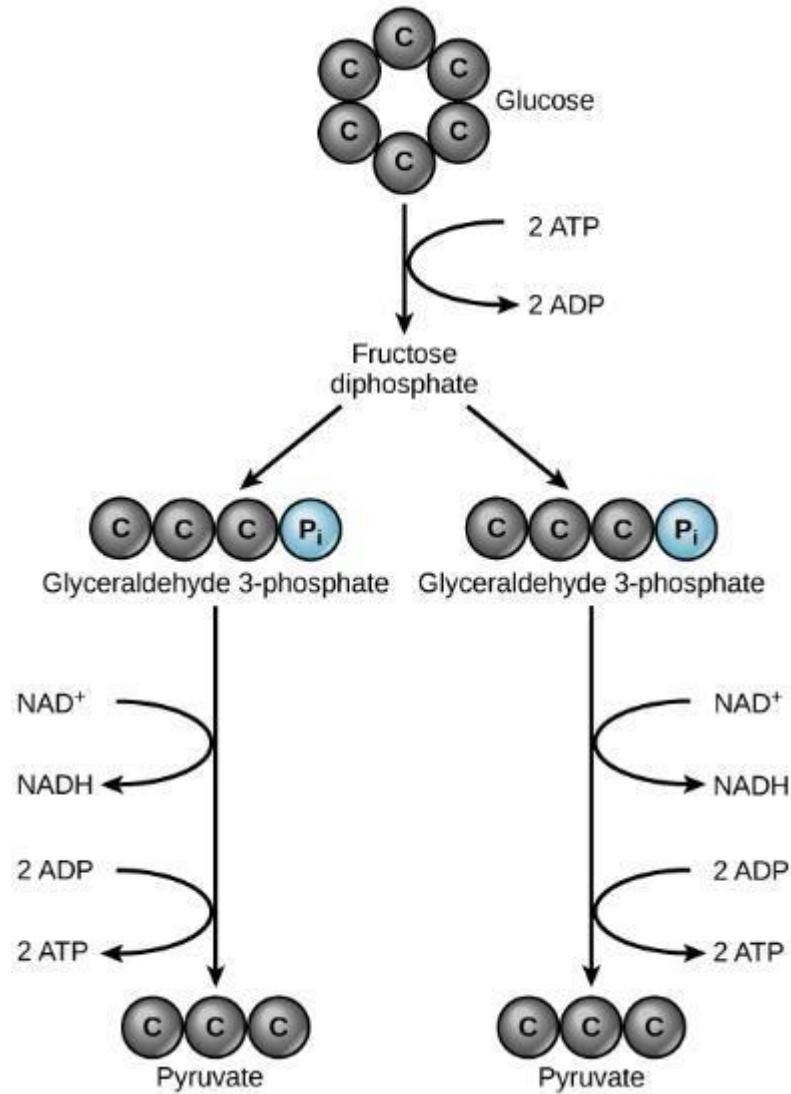


[Bing Videos:](#)
Calvin Cycle

Cellular respiration

- Steps:
 1. Glycolysis
 2. Pyruvate processing
 3. Citric cycle
 4. Oxidative phosphorylation

glycolysis



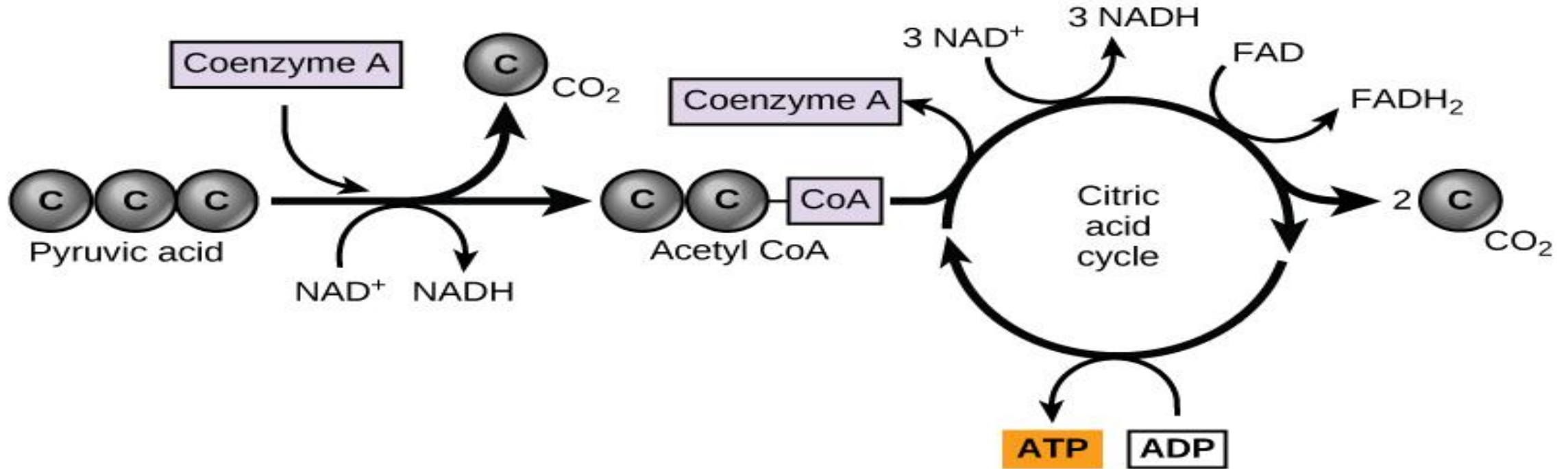
- In glycolysis, a glucose molecule is converted into two pyruvate molecules.
- 6 carbons-molecule (glucose) is transformed into 2 pyruvates (3-carbon-molecules)
- 1st phase – requires energy input
- 2nd phase – makes energy (ATP, NADH)

[Bing Videos](#)

Pyruvate processing & citric cycle

- EACH Pyruvate molecule is processed:
 - 1 carbon is lost as CO₂
 - 2 carbons are transformed into Acetyl-Coenzyme A
 - Makes energy as NADH
- Citric cycle:
 - Each carbon from Acetyl-CoA is lost as CO₂
 - Makes energy as ATP, NADH, FADH₂

pyruvate & Citric Cycle



- Pyruvate is converted into acetyl-CoA before entering the citric acid cycle.

[Bing Videos](#)